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A fast and reliable method for semi-automated planimetric quantification of dental plaque in clinical trials

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Abstract

Aim: To develop a simple and reproducible method for semi-automated planimetric quantification of dental plaque.

Materials and Methods: Plaque from 20 healthy volunteers was disclosed using erythrosine, and fluorescence images of the first incisors, first premolars, and first molars were recorded after 1, 7, and 14 days of de novo plaque formation. The planimetric plaque index (PPI) was determined using a semi-automated threshold-based image segmentation algorithm and compared with manually determined PPI and the Turesky modification of the Quigley–Hein plaque index (TM-QHPI). The decrease of tooth autofluorescence in plaque-covered areas was quantified as an index of plaque thickness (TI). Data were analysed by analysis of variance (ANOVA) and Pearson correlations.

Results: The high contrast between teeth, disclosed plaque, and soft tissues in fluorescence images allowed for a fast threshold-based image segmentation. Semi-automated PPI is strongly correlated with manual planimetry ($r = 0.92$; $p < .001$) and TM-QHPI recordings ($r = 0.88$; $p < .001$), and may exhibit a higher discriminatory power than TM-QHPI due to its continuous scale. TI values corresponded to optically perceived plaque thickness, and no differences were observed over time ($p > .05$, ANOVA).

Conclusions: The proposed semi-automated planimetric analysis based on fluorescence images is a simple and efficient method for dental plaque quantification in multiple images with reduced human input.

KEYWORDS
dental plaque index, digital image processing, light-induced fluorescence, planimetry

Clinical Relevance

Scientific rationale for study: Dental plaque accumulation is an important outcome of many clinical investigations. Planimetric analysis is a reliable but typically highly time-consuming method to quantify dental plaque, especially when used to assess multiple tooth surfaces in large-scale clinical trials.

Principal findings: Semi-automated planimetry reduced both the human input and the time required for planimetric quantification and may offer better discrimination in plaque accumulation than clinical indices.

Practical implications: The proposed method can be used for fast and reliable quantification of dental plaque in multiple images with reduced subjectivity compared to clinical plaque indices.

1 | INTRODUCTION

Dental plaque quantification methods can be used to evaluate the oral hygiene status of patients as well as to assess the efficacy of mechanical therapies or chemical agents for plaque control. Plaque indices are commonly used in clinical trials to quantify plaque accumulation on tooth surfaces. They involve visually scoring the extension or thickness of the plaque deposits using ordinal scales, with or without the aid of plaque-disclosing agents (Fischman, 1986). One of the most frequently used indices is the Turesky modification of the Quigley–Hein plaque index (TM-QHPI), which consists of pre-defined scores (from 0 to 5) that grade the coronal extension of the dental plaque from the gingival margin (Turesky et al., 1970). Because of their categorical nature, however, plaque indices may fail to detect potentially relevant changes in plaque accumulation. Scoring systems are also affected by the examiner's subjectivity, and extensive calibration and training of all assessors are required to ensure reliable recordings (Eaton et al., 1997).

Planimetric quantification of plaque overcomes many limitations of dental plaque indices. Planimetry uses image analysis to determine the relative area of the tooth covered by plaque, which is called the plaque percentage index or planimetric plaque index (PPI). PPI measures the extent of plaque coverage on a continuous scale and allows for a more objective, reliable, and precise quantification of dental plaque formation, compared to clinical indices (Quirynen et al., 1991; Söder et al., 1993). The method is normally applied to conventional digital photographs of disclosed plaque, in which areas of interest—the plaque deposits on the tooth—are determined and used to calculate PPI. For that, the images need to be segmented into different parts: the tooth area, the plaque area, and the background (soft tissues). Because of the low contrast and an overlap of colour spectra between teeth, gingiva, and dental plaque in white-light images, segmentation is usually performed manually. The image analysis is, therefore, highly time consuming, especially when used to assess multiple tooth surfaces in large-scale clinical investigations (Coulthwaite et al., 2009).

Fluorescence images of disclosed plaque provide a strong contrast between green autofluorescent tooth surfaces and plaque-covered areas visualized with an indicator dye. Therefore, they lend themselves to automated detection of plaque-covered areas based on intensity thresholding. Here, we present a simple, semi-automated method for planimetric plaque quantification using digital image analysis of fluorescence images after standard disclosing procedures. Planimetric measurements obtained with the newly developed method were compared with TM-QHPI recordings and with conventional planimetry. Additionally, we quantified the intensity decrease of tooth autofluorescence in areas covered by plaque as a measure for plaque thickness.

2 | MATERIALS AND METHODS

2.1 | Participants

Twenty healthy volunteers (12 males) aged 21–55 years (M_{age} = 27.4 years) with at least 20 natural teeth and no signs of periodontal disease or active carious lesions were enrolled in this study. Subjects with retainers, orthodontic appliances, intra-oral piercings, or prostheses were excluded. Pregnant or nursing women and individuals who had received anti-microbial or anti-inflammatory therapy within the past 30 days were also excluded. The study was approved by the Ethical Committee of Region Midtjylland (1-10-72-259-21) and conducted in accordance with the Helsinki Declaration and its amendments. The project was registered at the internal database for research projects at Aarhus University (2021-0214784; 24 June 2021). Written informed consent was obtained from all participants.

2.2 | Clinical procedures and image acquisition

After an initial professional tooth-cleaning, the participants were instructed to refrain from any oral hygiene measures, including mouthwashes, for 14 days. The facial aspects of the central incisor, first premolar, and first molar on day 1 (first quadrant), day 7 (second quadrant), and day 14 (third quadrant) were imaged using a VistaCam iX HD Smart intra-oral camera (Dürr Dental, Bietigheim-Bissingen, Germany) coupled with a fluorescence camera head.

First, the plaque was disclosed by gently pressing a foam pellet saturated with a red disclosing dye (5% erythrosine; Top Dent Rondell Röd, Top Dent Lifco Dental AB, Enköping, Sweden) against the tooth surfaces. Thereafter, the participants were asked to rinse with water for 10 s to remove the unbound dye, teeth were air-dried for 3 s, and fluorescence images of the disclosed plaque were acquired. At the end of each clinical session, the disclosed quadrant was professionally cleaned.

All images were obtained with dimmed room lights to reduce interference from ambient light and immediately after air-drying the tooth surface. Imaging antagonist teeth was avoided by acquiring the images with sufficient mouth opening. The camera was positioned at a standardized distance with 90° angulation from the tooth surface using a custom-made black spacer adapted to the camera head. The fluorescence images were displayed using the plaque detection setting of VistaCam's dedicated imaging software (DBSWIN 5.17.0, Dürr Dental) and exported as tif files. The camera settings are presented in Table S1. All image files were pseudonymized before analysis.

Additionally, one orthodontic patient was selected from among those referred for treatment to the Department of Dentistry and Oral Health (Aarhus University, Denmark). A set of images of the disclosed plaque around the orthodontic brackets was acquired in fluorescence mode as previously described.

2.3 | Plaque index

TM-QHPI was recorded for all fluorescence images of dental plaque. Scores from 0 to 5 were assigned to each tooth, as described in Table S2. Two examiners (Y.C.D.R. and K.K.J.) were initially trained and calibrated using a set of 25 photographs of disclosed plaque, scored twice with at least a 1-week interval between sessions. The calibrated examiners (Y.C.D.R. and K.K.J.) independently scored the plaque index of the entire set of images. Disagreements were resolved by a third examiner (S.S.).

2.4 | Digital image analysis

Red (disclosed plaque) and green (autofluorescent teeth) channel images were imported into the software daime (digital image analysis in microbial ecology, version 2.2.2) (Daims et al., 2006). For planimetric analysis, the merged images (red and green channels) were segmented based on intensity thresholding, which identified the teeth (*T*; clean and plaque-covered areas) as objects. The resulting object layer was then transferred to the red channel images and the contrast slightly enhanced (1.5×). Then the green channel images were subtracted from the red channel images, which left only the plaque-covered areas (*R*) in the images. In a last threshold-based segmentation step, the plaque-covered areas were identified as objects, and PPI was calculated for each image according to Equation (1):

$$\mathrm{PPI} = \frac{R}{T} \times 100. \quad (1)$$

In a subset of six participants (54 images), the tooth areas covered by plaque were manually selected using the object selection tool of the software daime and used to calculate the PPI. The processing times for manual planimetry and semi-automated planimetry were determined in a randomly chosen set of 10 images.

In plaque-covered areas, the green autofluorescence of the teeth was partly masked by the red dye. We quantified the reduction in green fluorescence intensity in those areas as a measure for plaque thickness. Object masks were extracted from the segmented merged (red and green) images (*T*) and the segmented red channel images (*R*). Then the red channel image masks were subtracted from the merged image masks, which resulted in a new mask comprising only the clean (not plaque-covered) areas of the teeth (*C*). The object masks for the plaque-covered (*R*) and the clean (*C*) areas were transferred to the green channel images, and the average fluorescence intensities were determined in both clean (IC) and plaque-covered (IR) areas. The plaque thickness index (TI) was calculated according to Equation (2):

$$\mathrm{TI} = 100 - \left(\frac{\mathrm{IR}}{\mathrm{IC}} \times 100\right). \quad (2)$$

The image analysis procedure is illustrated in Figure 1.

In a subset of 10 randomly chosen fluorescence and the corresponding white-light images, the colour and brightness contrast between plaque-covered areas, clean tooth areas, and surrounding soft tissues were determined (Supplementary Method S1).

2.5 | Intra- and inter-operator reproducibility of semi-automated planimetry

Fluorescence images of disclosed plaque ($N = 25$ teeth) were independently acquired by two investigators (Y.C.D.R. and K.K.J.) according to the image acquisition protocol previously detailed. Both operators acquired two consecutive sets of images of all teeth. PPI was quantified by semi-automated planimetry, and the intra- and inter-operator reproducibility was determined by intra-class correlation coefficients (ICCs).

2.6 | Statistical analysis

Normal distribution of the means was assessed with quantile–quantile plots and Shapiro–Wilk normality tests. Linear correlations between manual and semi-automated PPI and between PPI and TM-QHPI scores were assessed with Pearson correlation coefficients (ρ). Intra- and inter-operator reproducibility of semi-automated planimetry was determined by calculating the ICCs for absolute agreement using a two-way mixed-effects model and a two-way random-effects model, respectively. TI values were compared over time by analysis of variance (ANOVA). Pairwise comparisons of image contrast between fluorescence images and the corresponding white-light images were performed by paired *t*-tests. The effect sizes for the paired differences were calculated using Cohen's *d*. All analyses were performed using the software R (www.r-project.org). The significance level (α) was set at .05.

3 | RESULTS

All participants completed the study without deviations from the protocol. Compared to conventional white-light images, images recorded in fluorescence mode considerably increased the brightness contrast between the tooth and the surrounding tissues (36.70 ± 18.50 SD

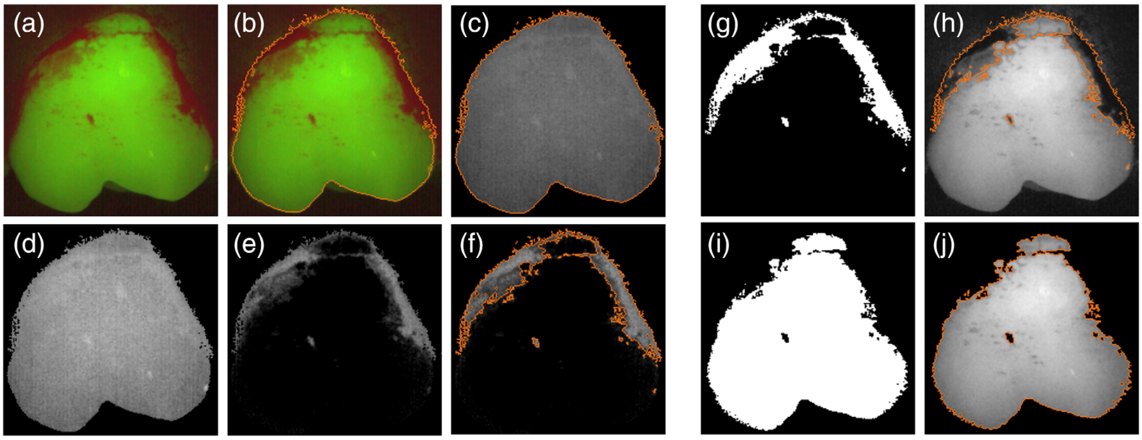

FIGURE 1 Digital image analysis procedure. (a–f) Calculation of the planimetric plaque index (PPI). To calculate the PPI, merged images (red and green channels, a) were segmented based on intensity thresholding (b) to determine the area of the entire tooth (indicated with an orange line). The object layer of those images was transferred to the red channel images (c), and the contrast was slightly enhanced (1.5×; d). Then the green channel images were subtracted from the red channel images, leaving only plaque-covered areas (e). The plaque-covered areas were identified as objects by intensity thresholding (f, orange line), and the PPI was calculated as the plaque-covered area divided by the total tooth area, multiplied by 100 (18.1% for the shown example). (g–j) Calculation of the reduction in green fluorescence intensity as a measure for plaque thickness. Object masks were created for plaque-covered areas (g) and clean tooth areas (i), and the masks were transferred to the green channel images (h, j). Average fluorescence intensities were determined for the plaque-covered areas (orange line in h) and the clean tooth areas (orange line in j), and the reduction of green fluorescence in plaque-covered areas was calculated (65.2% in the shown example).

vs. 65.10 ± 8.40 SD; $p < .001$, Cohen's $d = 1.78$). Fluorescence images also exhibited a significantly higher red/green colour contrast between plaque-covered and clean tooth areas compared to white-light images (91.90 ± 18.65 SD vs. 35.10 ± 8.41 SD; $p < .001$, Cohen's $d = 2.44$) (Figure S1). These properties allowed the rapid identification of plaque-covered tooth areas using digital image analysis (Figure 1). Semi-automated threshold-based image segmentation resulted in a 10-fold decrease in image processing time ($N = 10$ images) compared to manual delimitation of plaque-covered tooth areas (3 vs. 30 min).

Intra- and inter-operator reproducibility of semi-automated planimetry was excellent, with ICC values of 0.99 (95% confidence interval [CI] 0.98–0.99) for both operators (Y.C.D.R. and K.K.J.) and 0.98 (95% CI 0.96–0.99) for the inter-operator reproducibility. PPI scores determined by semi-automated image processing showed a strong correlation with manually determined PPI ($r = 0.92$; $p < .001$). Mean PPI (±SD) was 23.27% (11.95%) after 1 day (first quadrant), 59.54% (14.22%) after 7 days (second quadrant), and 36.22% (15.62%) after 14 days (third quadrant). Mean TM-QHPI scores (±SD) were 2.43 (±0.43), 4.08 (±0.63), and 3.52 (±0.46) after 1, 7, and 14 days, respectively. PPI determined by semi-automated image analysis was highly correlated with TM-QHPI recordings ($r = 0.88$; $p < .001$) but exhibited a better discriminative power to detect differences in plaque accumulation, especially for TM-QHPI scores 3–5 (Figure 2). The proposed method was also able to quantify plaque around orthodontic brackets (Figure 3).

Plaque TI values corresponded well with optically perceived plaque thickness (Figure 4). As both young and old plaque exhibited thick and thin areas, the mean TI (±SD) was not significantly different between days 1 (56.25 ± 10.88), 7 (57.78 ± 8.56), and 14 (54.88 ± 7.63) ($p > .05$; ANOVA).

4 | DISCUSSION

We developed a simple and reproducible digital image analysis procedure for semi-automated planimetric quantification of disclosed plaque using fluorescence images. Quantification relied on standard chairside plaque disclosing procedures, a commercially available intraoral fluorescence camera, and a freeware for image analysis. As a proof of concept, semi-automated PPI recordings were compared with traditional planimetry based on manual identification of plaque-covered areas during 2 weeks of de novo dental plaque formation. Both methods showed very good agreement, but as there is no gold standard (i.e., histology) for the exact quantification of plaque deposits in clinical settings, both methods lack external validity. One may, however, argue that semi-automated planimetry, in addition to a considerably reduced processing time (Coulthwaite et al., 2009), offers more objectivity during plaque quantification. Depending on the local plaque thickness, disclosing procedures result in different staining intensities, and oftentimes there is no clear cut-off but a gradual transition between plaque-covered and clean areas. During conventional planimetry, individual regions of interest need to be determined subjectively by the investigator for each image (Söder et al., 1993; Staudt et al., 2001; Brent Bowen et al., 2015). In contrast, cut-off thresholds for semi-automated planimetry are defined once by the operator, after which image segmentation is performed without additional human input.

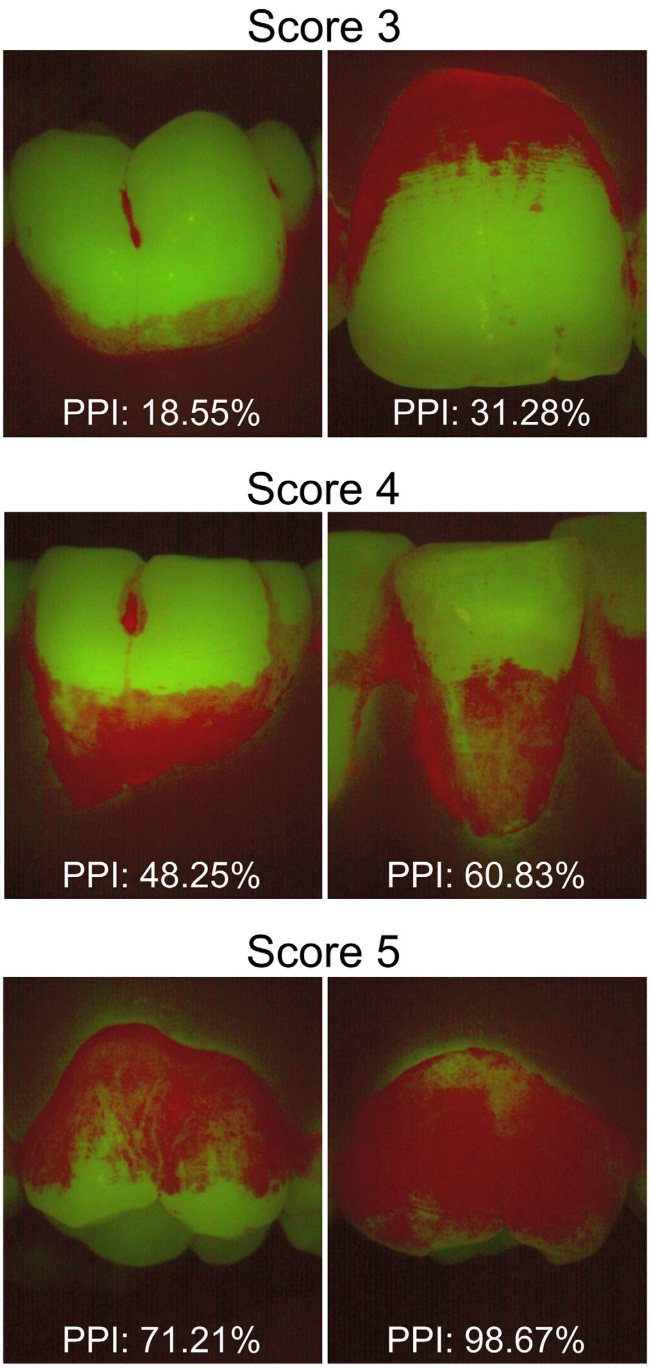

FIGURE 2 Discriminatory power of the planimetric plaque index (PPI) and the Turesky-modification of the Quigley–Hein plaque index (TM-QHPI). PPI exhibited a higher discriminatory power compared with TM-QHPI to detect differences in plaque accumulation, in particular for scores 3–5. The displayed examples illustrate how PPI may differ for identical TM-QHPI scores.

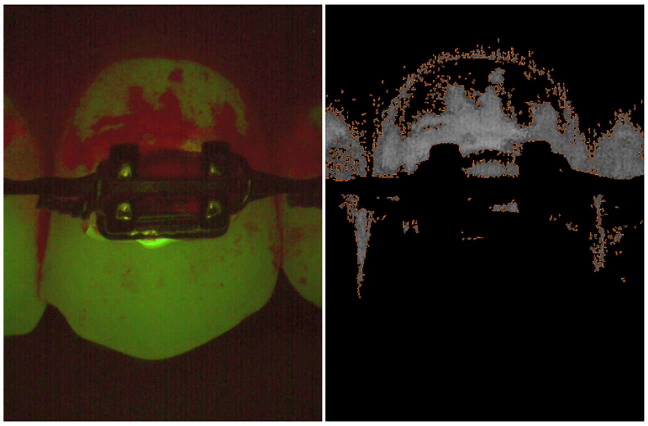

FIGURE 3 Semi-automated planimetric plaque quantification around orthodontic brackets. The proposed digital image analysis procedure allows the quantification of dental plaque around orthodontic brackets. The non-fluorescent brackets are excluded from both the plaque-covered and the clean tooth area, and the plaque percentage index is calculated for the remaining tooth surface (34.15% in the shown example).

Semi-automated image processing with the proposed algorithm was made possible using a fluorescence camera for image acquisition. On conventional white-light photographs of disclosed plaque, the spectral overlap between the colours of teeth, soft tissues, and dental plaque limits the application of threshold-based approaches. Fluorescence images enhance the contrast between soft tissues (non-fluorescent), teeth (green autofluorescent), and the red disclosed plaque (Pretty et al., 2004, 2005; Han et al., 2016), and thereby provide a better basis for semi-automated image segmentation.

Accurate image acquisition with a standardized set-up is of crucial importance for reliable identification of plaque-covered areas in fluorescence images. Artefacts caused by the interference of environmental light, the autofluorescence of antagonist teeth, or reflections from saliva-covered surfaces may limit the accuracy of PPI measurements. In this study, all fluorescence images were acquired under dimmed room lights using a custom-made black spacer to prevent artefacts caused by external visible light. Teeth were carefully air-dried prior to image acquisition, and imaging of antagonist teeth was avoided by sufficient mouth opening.

Some previous studies have used fluorescence images for PPI measurements (Coulthwaite et al., 2009; Ganss et al., 2020; Klaus et al., 2020), but plaque areas were typically determined manually using drawing tools in image editing software. A previously published semi-automated approach relied on the use of fluorescein, a pH-sensitive fluorescent dye that is excited by ultraviolet (UV) light (Sagel et al., 2000; Klukowska et al., 2011). Although the method produced reliable recordings, it could not be performed chairside and required the repeated application of buffered rinses, as the fluorescence emission of fluorescein declines at low pH (Martin & Lindqvist, 1975). Mansoor et al. (2015) proposed a semi-automated method to detect and quantify undisclosed autofluorescent plaque using a dedicated software developed by them (Mansoor et al., 2015). The software, however, is not publicly available and the quantification of autofluorescent plaque significantly underestimates the total plaque accumulation, since not all oral bacteria exhibit autofluorescence (Ganss et al., 2020; Klaus et al., 2020).

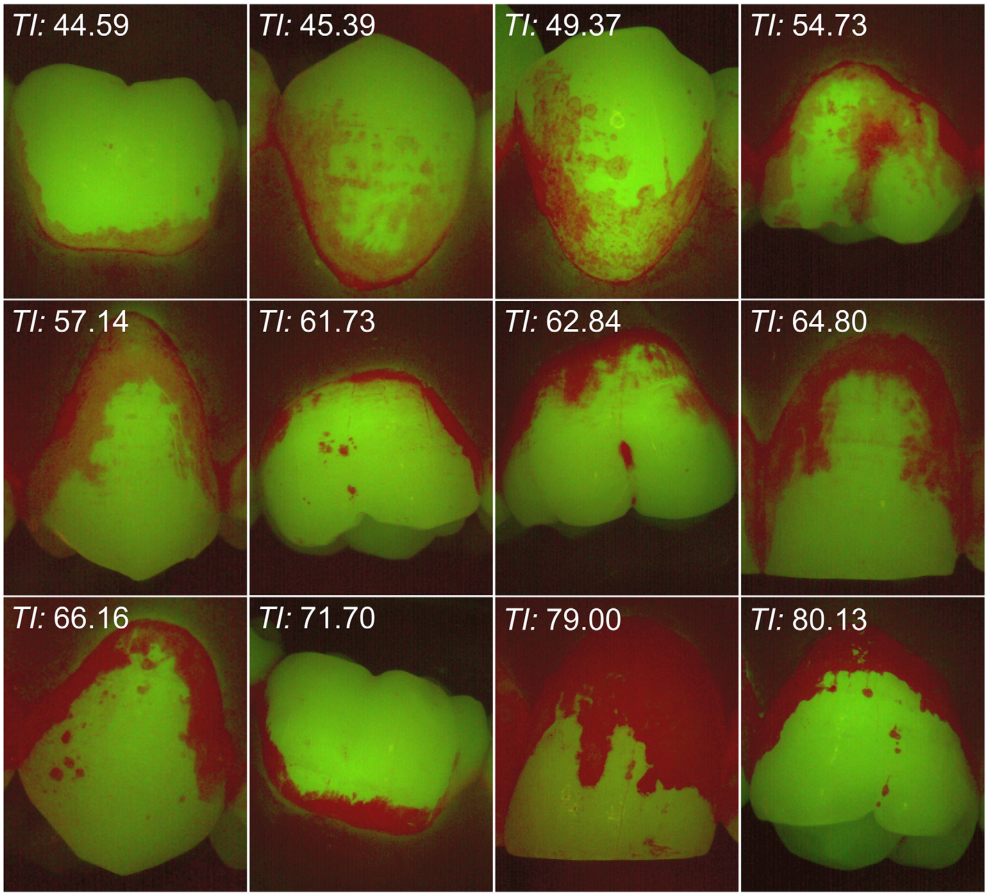

FIGURE 4 Representative disclosed dental plaque images and their respective thickness indices (TI). When covered with plaque, the green autofluorescence of the tooth was reduced on the fluorescence images, and the extent of the reduction corresponded visually to plaque thickness. The information may be used to calculate a plaque TI.

The present study showed that PPI measurements correlated well with TM-QHPI recordings. Owing to its continuous scale, PPI may exhibit a higher discriminatory power to detect differences in total plaque accumulation (Söder et al., 1993), but it remains to be determined if PPI differences that lie within one TM-QHPI score are clinically relevant. Moreover, planimetry does not discriminate between plaque deposits in different tooth areas. Several clinical indices have been specifically developed to assess the level of plaque accumulation along the gumline (Silness & Löe, 1964; Turesky et al., 1970; Rustogi et al., 1992), in the approximal space (Lange et al., 1977), or around orthodontic brackets (Kiliçoğlu et al., 1997; Beberhold et al., 2012), and they may remain the method of choice for focused plaque quantification in those sites. Alternatively, semi-automated planimetry could be refined by introducing pre-determined regions of interest for specific tooth areas. The presented semi-automated planimetric method may also be used for plaque quantification on lingual tooth surfaces, but the positioning of the camera and hence the standardized acquisition of images may be hampered in some regions of the mouth. Moreover, the method does not allow for PPI measurements on approximal surfaces, but these limitations apply to both manual and semi-automated image analysis procedures. Compared to traditional planimetry, the equipment cost for semi-automated planimetry is higher (3D printer, fluorescence camera), but the initial outlay may be outweighed by the reduced working hours during analysis.

While PPI measurements offer a precise estimate of plaque area coverage (Söder et al., 1993), they fail to identify changes in plaque thickness, which are highly related to pathogenicity (Larsen & Fiehn, 2017). To date, there is no validated method to assess dental plaque thickness in vivo. In fluorescence images, plaque-free tooth surfaces exhibit a bright green fluorescence, which is partially masked by the red disclosed plaque. We proposed a simple TI based on the green fluorescence intensity reduction as a measure for plaque thickness. TI could clearly discriminate between plaque that was visually perceived as thick or thin. As the analysis does not require additional equipment, the proposed index may represent a convenient and simple method to compare plaque thickness in clinical trials. However, the actual correlation between *TI* and plaque thickness remains to be validated.

5 | CONCLUSION

In conclusion, planimetric analysis is a reliable but typically time-consuming method for plaque quantification. In this study, a semi-automated DIA procedure for planimetric quantification was developed using standard disclosing procedures and fluorescence images, which allowed for threshold-based image segmentation due to the enhanced contrast between tooth, plaque, and soft tissue areas. The semi-automated analysis reduced the human input and the time required for planimetric quantification. The proposed method can be used for fast and reliable assessment of dental plaque in clinical trials, and may be further adapted according to the purpose of the investigation.

AUTHOR CONTRIBUTIONS

Yumi Chokyu Del Rey: Acquisition of data; statistical analysis; interpretation of data; manuscript preparation. **Pernille Dukanovic Rikvold**: Conception and design of the study; acquisition of data. **Karina Kambourakis Johnsen**: Acquisition of data. **Sebastian Schlafer**: Conception and design of the study; statistical analysis; interpretation of data; manuscript preparation. All authors critically reviewed the manuscript and gave final approval and are accountable for all aspects of the work.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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